

## Dividing Polynomials (2.4)

Name \_\_\_\_\_

### Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER:  $x + 4$

# \_\_\_\_\_ Find the remainder when  $f(x) = x^2 - 4$  is divided by  $(x - 3)$ .

ANSWER: 17

# \_\_\_\_\_ List the possible rational zeros of  $f(x) = x^2 - 2$ .

ANSWER: 0

# \_\_\_\_\_ Find the remainder when  $f(x) = x^3 - 2x + 1$  is divided by  $(x + 1)$ .

ANSWER: yes

# \_\_\_\_\_ Divide  $x^2 - 9$  by  $(x - 3)$ ; give the quotient.

|                                                                                                                                                       |                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <p>ANSWER: 5</p> <p># _____ By the Factor Theorem, <math>(x - k)</math> is a factor of <math>f</math> exactly when <math>f(k)</math> equals what?</p> | <p>ANSWER: 1</p> <p># _____ Is <math>(x - 2)</math> a factor of <math>f(x) = x^3 - 8</math>?</p>                                     |
| <p>ANSWER: <math>2x + 5</math></p> <p># _____ For <math>f(x) = 2x^3 - x + 3</math>, find <math>f(2)</math>.</p>                                       | <p>ANSWER: <math>x + 3</math></p> <p># _____ Divide <math>x^2 + 7x + 12</math> by <math>(x + 3)</math>; give the quotient.</p>       |
| <p>ANSWER: 2</p> <p># _____ Divide <math>2x^2 + 3x - 5</math> by <math>(x - 1)</math>; give the quotient.</p>                                         | <p>ANSWER: <math>\pm 1, \pm 2</math></p> <p># _____ Find the remainder when <math>x^4</math> is divided by <math>(x - 1)</math>.</p> |